

The Datasaurus Dozen

Why Visualization Matters in Data Science

Demo Class for Level 4 Undergraduate Students

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November 20, 2025

Can You Trust Summary Statistics Alone?

Quick Question

If I told you two datasets have:

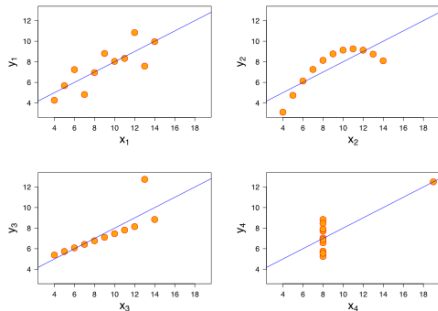
- Same mean: $\bar{x} = 54.26$, $\bar{y} = 47.83$
- Same standard deviation: $\sigma_x = 16.76$, $\sigma_y = 26.93$
- Same correlation: $r = -0.06$

Would you say they're basically the same data?

The Classical Example: Anscombe's Quartet (1973)

The Setup:

- Four datasets by Francis Anscombe
- **ALL** have identical statistics:
 - Mean of $x = 9.0$
 - Mean of $y = 7.5$
 - Variance of $x = 11.0$
 - Correlation $r = 0.816$



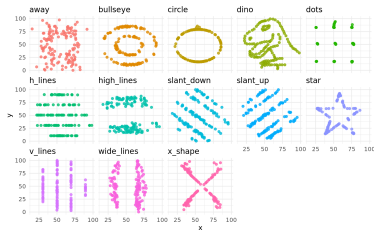
What the Plots Reveal

Dataset	Pattern
I	Perfect linear relationship
II	Clear non-linear (quadratic) pattern
III	Perfect linear except one outlier
IV	No relationship except one influential point

Enter the Datasaurus (2017)

Making it Even More Dramatic:

- Autodesk researchers asked:
"Can we make this more surprising?"
- Created **13 datasets** with:
 - Nearly identical summary statistics
 - Wildly different visual patterns
- Used **simulated annealing** algorithm
- Morphed shapes while preserving statistics



Watch: Dinosaur → Star → Circle (statistics stay constant!)

The Datasaurus Dozen: Identical Statistics

Dataset	$\bar{x}$	$\bar{y}$	σ_x	σ_y	r
Dinosaur	54.26	47.83	16.76	26.93	-0.06
Star	54.26	47.84	16.76	26.93	-0.06
Circle	54.26	47.84	16.76	26.93	-0.06
Bullseye	54.26	47.83	16.76	26.93	-0.06
X-shape	54.26	47.84	16.76	26.93	-0.06
Lines	54.26	47.84	16.76	26.93	-0.06
Dots	54.26	47.84	16.76	26.93	-0.06

Why This Matters: Healthcare Data

Scenario: Analyzing Patient Recovery Times

Comparing two treatments for the same condition

Summary Statistics Say:

- Treatment A: Mean = 14 days
- Treatment B: Mean = 14 days
- p -value = 0.45
- “No significant difference”

Visualization Reveals:

- Treatment A: Consistent ~ 14 days
- Treatment B: Bimodal!
 - Either 7 days (fast)
 - Or 21 days (slow)

Why This Matters: Business Analytics

Scenario: E-Commerce Customer Spending

Analyzing purchasing behavior for marketing strategy

Statistics:

- Mean spend = £50
- Median spend = £48
- Std dev = £25

Looks normal, right?

Histogram Shows:

- Most: £10-£20
- Few high-value: £500+
- **Heavily right-skewed**

Completely different!

Why This Matters: Environmental Data

Scenario: Temperature Readings from Weather Stations

Two stations with identical annual statistics

Summary Statistics:

- Station A: Mean = 15°C
- Station B: Mean = 15°C
- Both: Variance = 8

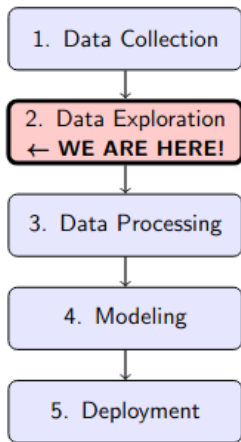
Same climate?

Time Series Plot:

- Station A: Steady 15°C
(Tropical island)
- Station B: -5°C to 35°C
(Continental)

Vastly different!

The Data Science Pipeline



Exploratory Data Analysis (EDA) is Not Optional

✓ What EDA Catches:

- Non-linear relationships
- Outliers and anomalies
- Data entry errors
- Missing data patterns
- Multimodal distributions
- Heteroscedasticity
- Clustering in data

⚙️ Impact on Modeling:

- Choose appropriate models
- Transform variables correctly
- Handle outliers properly
- Validate assumptions
- Improve predictions
- Avoid false conclusions

! Example 1: Detecting Algorithmic Bias

Scenario: Your AI model is “95% accurate”

- Overall accuracy = 95%
- **But visualization reveals:**
 - Group A (majority): 99% accuracy
 - Group B (minority): 75% accuracy

This is how we discover bias before deployment!

The Ethical Imperative

Real-World Consequences

Blind trust in summary statistics can lead to:

- **Biased hiring algorithms** (discriminating against groups)
- **Unfair loan decisions** (redlining by proxy)
- **Misdiagnosed patients** (missing critical patterns)
- **Wrongful convictions** (flawed forensic analysis)

Always ask: "What am I not seeing in the numbers alone?"

Practical Tools & Best Practices

</> Tools You'll Use:

- **Python:**
 - matplotlib, seaborn, plotly
- **R:** ggplot2
- **Business:** Tableau, Power BI
- **Even Excel!**

The Golden Rules:

- 1 Always plot before analysis
- 2 Use multiple viz types
- 3 Look for outliers, patterns
- 4 Show data, not just stats

Minimum Viable EDA

For every dataset: **scatter plot + histogram + box plot**

Quick Demo: Python Code

```
import matplotlib.pyplot as plt
import pandas as pd
raw_url = "https://raw.githubusercontent.com/jsanchez/DemoClass/
          main/data/raw/datasaurus_dozen.csv"
data = pd.read_csv(raw_url)
# Plot all 13 datasets
fig, axes = plt.subplots(3, 5, figsize=(15, 9))
datasets = df['dataset'].unique()
for ax, dataset in zip(axes.flatten(), datasets):
    subset = data[data['dataset'] == dataset]
    ax.scatter(subset['x'], subset['y'], alpha=0.5)
    correlation_value = subset[['x', 'y']].corr().iloc[0,1]
    ax.set_title(f'{dataset}\nr2=2{correlation_value:.2f}')
plt.tight_layout()
plt.show()
```

Three Key Takeaways

1. Statistics Without Visualization

Summary statistics are powerful but **incomplete**. Always visualize.

2. This Applies to EVERY Stage

From cleaning → validation → deployment
Visualization is your **safety check** at each step.

Resources for Further Learning

Original Research

- Matejka & Fitzmaurice (2017) “Same Stats, Different Graphs”
- Interactive:
`autodeskresearch.com/publications/samestats`
- GitHub:
`AutodeskResearch/same-stats-different-graphs`

*Francis Anscombe created his quartet in 1973
to make this point.*

*50+ years later, data scientists
STILL make this mistake.*

Don't be one of them.

Always visualize. Always question. Always explore.

Interactive Challenge

? Can You Spot the Difference?

Two mystery datasets with identical statistics:

- Mean $X = 10.0$, Mean $Y = 8.0$
- Std Dev $X = 3.0$, Std Dev $Y = 2.5$
- Correlation $r = 0.75$

Question: Which would you trust more for prediction?